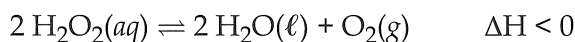


1.

2.2.9 (2016:01)

The decomposition of hydrogen peroxide in a closed system is represented by the equation below.



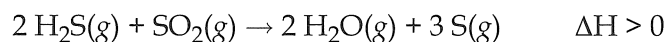
Which one of the following will increase the equilibrium yield of oxygen?

- (a) decreasing the concentration of hydrogen peroxide
- (b) increasing the total pressure of the system
- (c) decreasing the temperature of the system
- (d) adding an inert gas to the system

2.

2.2.1 (2016:02)

Sulfur can be obtained from hydrogen sulfide found in natural gas according to the equation below.



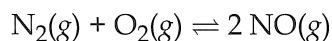
Which one of the following changes will initially decrease the rate at which sulfur is produced?

- (a) reduce the partial pressure of the hydrogen sulfide ($\text{H}_2\text{S}(g)$)
- (b) increase the partial pressure of sulfur dioxide ($\text{SO}_2(g)$)
- (c) add a metal catalyst to the reaction vessel
- (d) heating the reaction vessel

3.

2.2.11 (2016:03)

Consider the equilibrium system below.



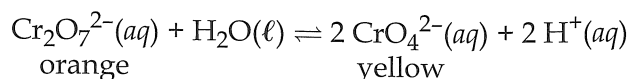
If the equilibrium constant (K) for this reaction is 4.1×10^{-31} , which one of the following statements is **true** for the system where the initial partial pressures of nitrogen and oxygen were equal to each other?

- (a) Once equilibrium is reached, the reverse reaction rate is much faster than the forward reaction rate.
- (b) The partial pressure of $\text{NO}(g)$ is less than the partial pressure of $\text{N}_2(g)$.
- (c) The actual ratio of gaseous N_2 particles to NO gaseous particles is 1:2.
- (d) When nitrogen gas is injected into a vessel containing mostly oxygen gas, the partial pressure of oxygen decreases dramatically.

4.

2.2.9 (2016:04)

A 500 mL solution of dichromate ions and chromate ions at equilibrium is described by the equation below.



Which of the following **best** describes the effect of adding 10 mL of concentrated potassium hydroxide solution to the system once equilibrium has been re-established?

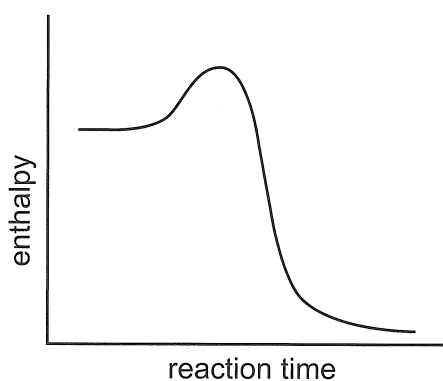
	Relative change in concentration of $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$	Relative change in concentration of $\text{CrO}_4^{2-}(\text{aq})$	Relative change in concentration of $\text{H}^+(\text{aq})$	Colour change of solution
(a)	increase	decrease	decrease	more orange
(b)	decrease	increase	decrease	more yellow
(c)	no change	no change	no change	no change
(d)	decrease	increase	increase	more yellow

5.

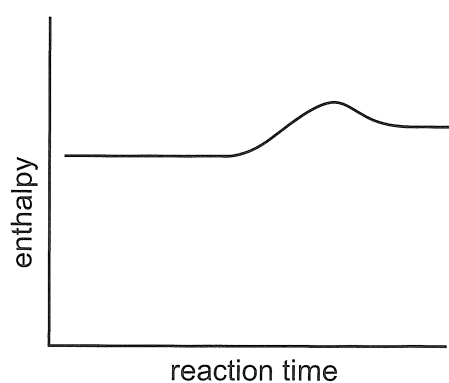
2.2.7 (2016:08)

Which of the following energy profile diagrams **best** represents a spontaneous, exothermic reaction?

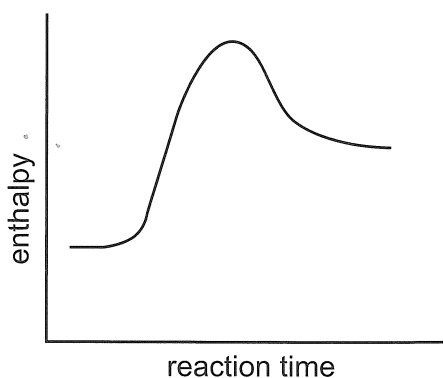
(a)



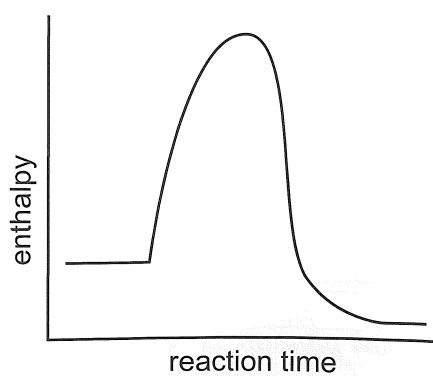
(b)



(c)



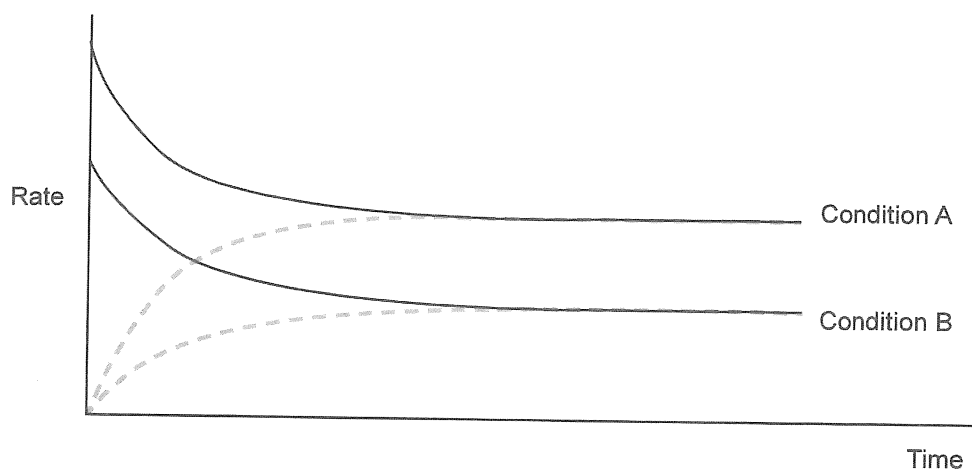
(d)



6.

2.2.1 (2017:05)

The following graph shows both the forward (—) and reverse (-----) reaction rates for a reaction under two different sets of conditions (Condition A and Condition B).



What is different about these conditions that would account for the graphs as shown?

- (a) Condition A is at a higher temperature.
- (b) Condition B is at a higher temperature.
- (c) Condition B had a catalyst added to it.
- (d) Condition B was conducted at a higher temperature and pressure.

7.

2.2.1 (2017:07)

Consider the following statements about the effect of a catalyst being added to a reaction mixture.

- (i) The formation of intermediate species not found in the uncatalysed reaction.
- (ii) The availability of a new reaction path having a lower activation energy.
- (iii) An increase in the percentage of collisions resulting in a reaction.
- (iv) The equal increase of both forward and reverse reaction rates.

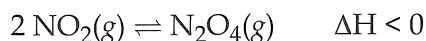
Which of these statements describe what will occur when a catalyst is added to a reaction mixture?

- (a) i and ii only
- (b) ii and iv only
- (c) i, ii and iv only
- (d) i, ii, iii and iv

8.

2.2.9 (2017:12)

A sealed glass tube at room temperature contains nitrogen dioxide (a brown gas) and dinitrogen tetroxide (a colourless gas) in equilibrium, as represented by the following equation.



If the appearance of the gas mixture at room temperature is pale brown, which one of the following is true if the glass tube is placed in hot water?

- (a) The gas mixture will not undergo any noticeable change in appearance.
- (b) The gas mixture will become darker brown.
- (c) The gas mixture will become even paler at first, but would then return to its original appearance of pale brown.
- (d) The gas mixture will become colourless.

9.

2.1 (2017:18)

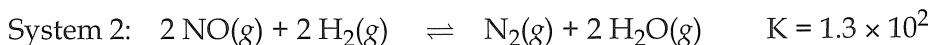
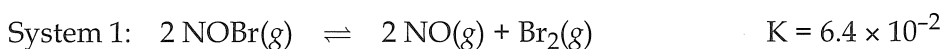
The acidification of oceans due to their increased concentrations of carbon dioxide decreases the rate and amount of calcification in some marine organisms, e.g. shellfish and coral reefs. Which one of the following equations best represents the chemistry involved in decreasing the rate and amount of calcification?

- (a) $2 \text{H}^+ + \text{CaCO}_3 \rightarrow \text{Ca}^{2+} + \text{H}_2\text{O} + \text{CO}_2$
- (b) $\text{CO}_2 + \text{H}_2\text{O} + \text{CO}_3^{2-} \rightarrow 2 \text{HCO}_3^-$
- (c) $4 \text{H}^+ + 2 \text{CO}_3^{2-} \rightarrow \text{H}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2$
- (d) $\text{CO}_2 + \text{Ca}(\text{OH})_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$

10.

2.2.11 (2018:02)

Consider the following reaction systems at equilibrium.



Which of these statements regarding these systems is/are true?

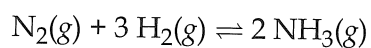
- (i) System 2 reaches equilibrium faster than System 1.
- (ii) The greatest ratio of products to reactants occurs in System 2.
- (iii) Equilibrium in System 1 favours the reactants more than it does in System 2.

- (a) i only
- (b) ii only
- (c) i and iii only
- (d) ii and iii only

11.

2.2.4 (2019:02)

The Haber Process involves the following equilibrium reaction:



A number of closed reaction vessels were set up containing the gases shown in the table below.

Reaction vessel	Gases initially present
i	nitrogen, hydrogen
ii	nitrogen
iii	ammonia
iv	hydrogen, ammonia

In which of the above closed reaction vessels would equilibrium be established after a period of time?

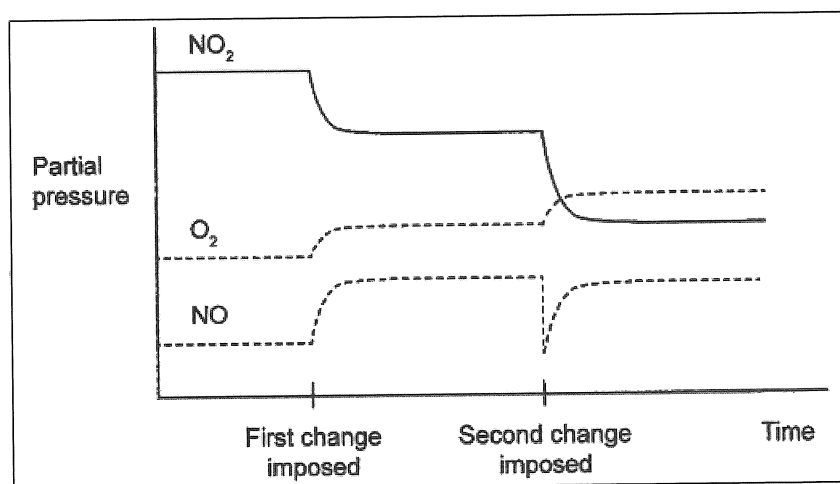
- (a) i only
- (b) i and iii only
- (c) i, iii and iv only
- (d) ii, iii and iv only

Questions (2019:03) and (2019:04) refer to the following information.

Nitrogen dioxide, $\text{NO}_2(\text{g})$, is formed when nitrogen monoxide, $\text{NO}(\text{g})$, undergoes oxidation as shown below.



A change was imposed on an equilibrium gas mixture of NO_2 , NO and O_2 . The mixture returned to equilibrium and another change was imposed. The following graph shows the effects of the two changes.



12.

2.2.9 (2019:03)

Identify the imposed changes that best account for the shape of the graph.

	First change	Second change
(a)	the temperature is decreased	the partial pressure of O_2 is increased
(b)	the temperature is decreased	the partial pressure of NO is decreased
(c)	the temperature is increased	the partial pressure of O_2 is increased
(d)	the temperature is increased	the partial pressure of NO is decreased

13.

2.2.11 (2019:04)

What do the initial partial pressures of the three gases indicate?

- (a) The relative proportions of the gases present at equilibrium.
- (b) That there is initially no NO gas present in the system.
- (c) That the NO_2 gas reaches equilibrium first.
- (d) That the O_2 and NO gases are producing NO_2 at a faster rate than they are being formed.

14.

2.1 (2019:12)

The United Nations Kyoto Protocol and the Intergovernmental Panel on Climate Change aim to secure a global commitment to reducing greenhouse gas emissions over the next few decades.

Which of the following equations shows the production of a greenhouse gas?

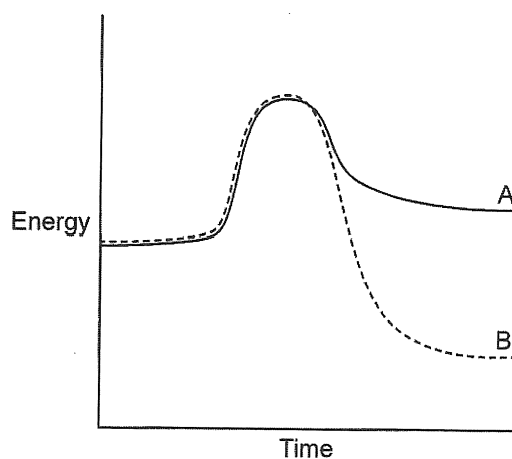
- (i) $\text{O}_2 + \text{O} \rightarrow \text{O}_3$
- (ii) $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$
- (iii) $\text{CH}_4 + 2 \text{O}_2 \rightarrow \text{CO}_2 + 2 \text{H}_2\text{O}$
- (iv) $\text{CO}_2 + 4 \text{H}_2 \rightarrow \text{CH}_4 + 2 \text{H}_2\text{O}$
- (v) $\text{NH}_4\text{NO}_3 \rightarrow 2 \text{H}_2\text{O} + \text{N}_2\text{O}$

- (a) i and ii only
- (b) ii and iii only
- (c) iii, iv and v only
- (d) i, ii, iii, iv and v

15.

2.2.6 (2020:08)

Energy profile diagrams for two different chemical reactions (A and B) are shown below.



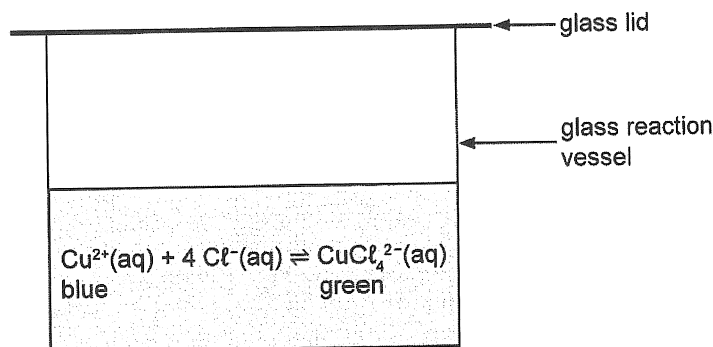
Which of these reactions is the more likely to be reversible and why?

- (a) Reaction A, because its forward reaction is endothermic.
- (b) Reaction B, because its forward reaction is exothermic.
- (c) Reaction A, because the activation energy of its reverse reaction is smaller than that for Reaction B.
- (d) Neither, because the activation energies of their forward reactions are the same.

16.

2.2.2 (2020:12)

A group of Chemistry students observed a demonstration in which solid sodium chloride was added to an aqueous solution of copper(II) chloride in a glass reaction vessel. A tightly-fitting lid was placed on the reaction vessel. The solid sodium chloride was allowed to dissolve and then the entire system was heated and then cooled. This resulted in colour changes. The reaction system is shown below.



The students were asked to decide if the system was open or closed and if the demonstration involved chemical and/or physical processes. Their responses are shown in the following table.

Which student has the correct responses?

	Student	Type of system (open/closed)	Type of process (chemical/physical)
(a)	1	closed	physical
(b)	2	closed	chemical
(c)	3	closed	chemical and physical
(d)	4	open	chemical

17.

2.2.4 (2020:16)

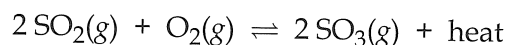
Which of the following is **not** a characteristic of a system in dynamic equilibrium?

- (a) The mass of the reactants equals the mass of the products.
- (b) Reactants are forming products and products are forming reactants.
- (c) The rates of the forward and reverse reactions are equal.
- (d) The position of the equilibrium is affected by temperature.

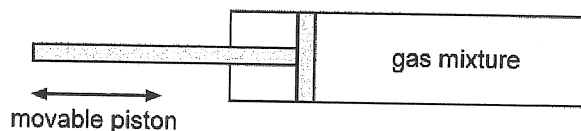
18.

2.2.11 (2020:24)

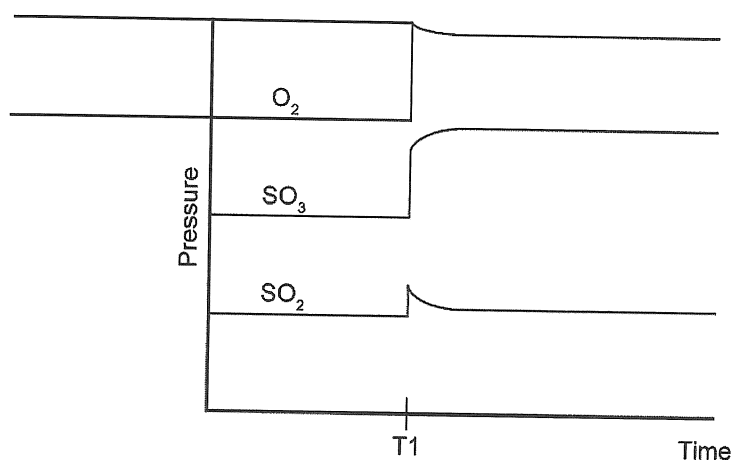
Consider the following equilibrium:



A mixture of these gases was at equilibrium in a sealed container with a movable piston, as shown below.



A change was applied to the system at T1 and the results of this change are shown in the following graph.



What was the change that occurred at T1?

- (a) An inert gas was added to the reaction vessel.
- (b) The reaction vessel was heated.
- (c) A catalyst was added to the reaction vessel.
- (d) The volume of the reaction vessel was decreased.

19.

2.1 (2021:15)

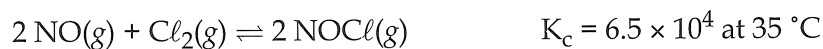
Which of the following processes does **not** contribute to the building of weaker seashells through ocean acidification?

- (a) $\text{HCO}_3^-(aq) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{CO}_3^{2-}(aq) + \text{H}_3\text{O}^+(aq)$
- (b) $2 \text{H}^+(aq) + \text{CaCO}_3(s) \rightleftharpoons \text{Ca}^{2+}(aq) + \text{CO}_2(aq) + \text{H}_2\text{O}(\ell)$
- (c) $\text{CO}_2(g) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{H}_2\text{CO}_3(aq)$
- (d) $\text{H}_2\text{CO}_3(aq) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{HCO}_3^-(aq) + \text{H}_3\text{O}^+(aq)$

20.

2.2.10 (2021:20)

Consider the following reversible reaction:



Which of the following statements describes the relative concentrations of reactants and products in this system when equilibrium is established in a closed vessel at 35°C ?

- (a) The concentrations of reactants and products will be equal.
- (b) There will be a greater concentration of products than reactants.
- (c) The reactant concentration will be greater than that of the products.
- (d) The concentrations of NOCl and NO will be double the concentration of Cl_2 .

21.

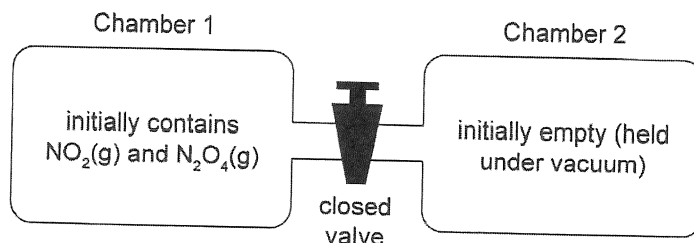
2.2.11 (2021:23)

The equilibrium position of a system depends on the concentrations of

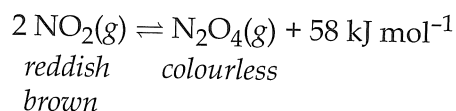
- (a) reactants only.
- (b) products only.
- (c) reactants and products.
- (d) neither reactants nor products.

Questions (2021:24) and (2021:25) refer to the following information.

Equal moles of nitrogen dioxide gas (NO_2) and dinitrogen tetroxide gas (N_2O_4) are sealed inside one half of a two-chamber reactor as shown below. The temperature inside both chambers is 25°C .



After a while, the following equilibrium is established inside Chamber 1:



22.

2.2.5 (2021:24)

More $\text{N}_2\text{O}_4(\text{g})$ is added to Chamber 1. What observations would be made about the colour of the gas mixture and its temperature after a new equilibrium is established?

	Gas mixture colour	Temperature
(a)	darker brown	higher
(b)	darker brown	lower
(c)	paler brown	higher
(d)	paler brown	lower

23.

2.2.9 (2021:25)

The valve between the chambers is opened, allowing the gas mixture in Chamber 1 to fill both chambers. Which statement describes the initial and final observations of the gas mixture's colour as a new equilibrium is established?

	Initial observation when tap is opened	Observation as equilibrium is re-established
(a)	paler brown	became darker
(b)	paler brown	became paler
(c)	darker brown	became darker
(d)	darker brown	became paler

Chapter 1: Chemical Equilibrium

- 1.(2016:01-c) The reaction is exothermic – cooling the system would give more oxygen as the reaction moves to the RHS. Adding inert gas would have no effect, as the partial pressure of oxygen does not change. Decreasing the conc of H_2O_2 would lead to a lower forward rate and less O_2 . Increasing the pressure would increase the partial pressure of oxygen which would result in less O_2 forming. The answer is 'c'.
- 2.(2016:02-a) Note the question asks for decrease in rate. Answers 'b', 'c' and 'd' always increase reaction rate. Decreasing the partial pressure of H_2S will decrease the frequency of collisions and thus the rate of formation of sulfur. The answer is 'a'.
- 3.(2016:03-b) Answer 'a' is wrong. At equilibrium the forward and reverse reaction rates are equal. Answer c would suggest a much larger K value. Answer d would be correct for a large K_c reaction. The value of K_c is very small. Thus $[\text{products}]/[\text{reactants}]$ is small i.e. there are few products. The answer is 'b'.
- 4.(2016:04-b) This is a classic question. If you are told something not in the equation is added, be sure it will probably react with something in the equation either as acid/base or a precipitate. The added solution is concentrated so there is no need to worry about diluting the reactants and products. Here the H^+ is destroyed so the equation moves to the right – the answer is 'b'.
- 5.(2016:08-a) The words spontaneous, exothermic means a reaction with a small E_a and a negative ΔH . The answer is 'a'.
- 6.(2017:05-a) The question is based on the fact that all reactions have a higher rate at higher temperatures. The answer is 'a'.
- 7.(2017:07-d) This question covers the multiple properties of catalysts that you should know. All are valid. The answer is 'd'.
- 8.(2017:12-b) Another LCP question. The addition of heat will favour the endothermic reaction which is the reverse reaction. The answer is 'b'.
- 9.(2017:18-b) This question is taken directly from the syllabus 'Acidification shifts the equilibrium of carbonate chemistry in seawater, decreasing the rate and amount of calcification'. Answer (b) shows the destruction of carbonate ions that are essential to the process of calcification. The answer is 'b'.
- 10.(2018:02-d) The magnitude of a K value indicates the product:reactant ratio. It says nothing about the rate of reaction. Therefore answer (i) is incorrect. Both (ii) and (iii) are true. The answer is 'd'.
- 11.(2019:02-c) In two places the question mentions that the system is closed. Therefore, equilibrium can be established if either
 1. all the reactants are present,
 2. all the products are present
 Therefore, item ii cannot establish equilibrium. The answer is 'c'.
- 12.(2019:03-d) At the first change the partial pressures slowly moved to new equilibrium positions indicating a new K value. Thus, it must have been a temperature change. As the reaction is exothermic and the partial pressure of NO_2 decreased the reaction must have been heated – answer 'c' or 'd'.
At the second point the partial pressure of NO decreased dramatically and then partly recovered, shown by (b) or (d). The answer is 'd'.
- 13.(2019:04-a) This question investigates the concept of equilibrium and could have been used to show that $K > 1$ for this reaction as at equilibrium the equation lies to the right-hand side. The question, however, as asked the answer is 'a'.
- 14.(2019:12-d) A greenhouse gas is any gas with more than two atoms. Some such as SF_6 are 23 000 times more effective than CO_2 but they are all greenhouse gases. Thus, these are all greenhouse gases – the answer is 'd'.
- 15.(2020:08) The magnitude of the reverse activation energy determines whether a reaction is reversible. Endothermic reactions have smaller reverse E_a than the forward E_a . The answer is (c).
- 16.(2020:12) A closed system allows energy to enter or leave but not matter. This is a closed system and by applying heat the equilibrium position was changing – this is a chemical process but it results in a colour change – a physical process. The answer is (c).
- 17.(2020:16) An equilibrium reaction does not go to completion so answer (a) is possible but unlikely. Answers (b) and (c) are correct as the reaction has not stopped and the forward and backward rates are equal. K_c is temperature dependent so (d) is correct. The answer is (a).
- 18.(2020:24) At T1 the partial pressure of each of the gases increased proportionally. This shows that the total volume decreased increasing the partial pressures. The reaction then moved to the right to overcome this constraint and re-establish equilibrium. The answer cannot be (a) as the partial pressures would not change, nor (b) as the reaction would move to favour one side, nor (c) as there would no change at T1 as the system was already at equilibrium. The answer is (d).
- 19.(2021:15) The sea-shell question concerns the formation of carbonate ions in sea water from geological sources and their conversion to hydrogen carbonate ions by hydrogen ions formed by the dissolution of carbon dioxide. Sea creatures cannot incorporate hydrogen carbonate ions into their shells. This is described in options a, c and d. The answer is b.
- 20.(2021:20) As K is such a large value it follows that $[\text{products}] \gg [\text{reactants}]$. The answer is b.
- 21.(2021:23) The equilibrium position described by K is determined only by $[\text{products}]$ and $[\text{reactants}]$. The answer is c.
- 22.(2021:24) Adding N_2O_4 will cause the system to move to the left to maintain K and so it will become more dark brown and will be colder as it uses heat. The answer is b.
- 23.(2021:25) When the second container is opened the gases occupy a greater volume so the colour will be less brown. The system will then move to re-establish K and will move to the side with more gas moles. It will become more brown. The answer is a.